Oral presentation script

FI510E AI & Business Applications, Midterm 1, Rabat Business School
Elisa Ouillet, Iulia Kozhina, Lucas Clodic | 7 minutes

Speaker	Slides	Part	Time
Elisa	1 to 3	Business problem and data	2 min
Iulia	4 to 6	Accuracy trap, models, final test	2 min
Lucas	7 to 10	Cost of errors, recommendation, web page, limits	2 min 45

- Speak slowly and look at the audience, not at the slides. The highlighted numbers are the ones to remember.
- If you run out of time, skip the ROC-AUC sentence (slide 6) and the 2.3% sentence (slide 7).
- Each speaker starts on a new page when printed.

Elisa slides 1 to 3, about 2 minutes

Slide 1. Title

Good morning. We are Elisa, Iulia and Lucas. Our project answers a simple question: if a bank can only call a few of its credit card clients this month, who should it call first?

Slide 2. The business problem

A credit card company has **4,500 clients**. Every month, about **22%** of them, so one in five, will miss their next payment. But the team only has time for **450 calls**, which is 10% of the clients.

The decision happens right after the September statement. The team calls before the October due date, with a reminder or an instalment plan. A call to the right person recovers part of the debt. A call to a good client wastes time and annoys them.

So our model has one job: put the clients who will not pay at the top of the list.

Slide 3. The data

We used a public dataset from the UCI repository: **30,000 clients** of a bank in Taiwan, in 2005. For each client we have the credit limit, the education level and six months of history: payment status, bill and amount paid. We added seven ratios, like the share of the credit limit used.

There are no missing values. We removed sex, marital status and age, because a bank should not use them in a credit decision, and it costs almost nothing. All the variables are known before the payment, so there is no leakage. Finally, we split the clients: 70% to train, 15% to choose the model, and 15% to test it only once, at the end.

Iulia will now show you the models.

Iulia slides 4 to 6, about 2 minutes

Slide 4. The accuracy trap

Before comparing models, one warning. The simplest rule is to say "nobody defaults". On our validation clients, this rule is right **78% of the time**. That sounds good, but look at the table: it finds **zero defaulters**. All 995 are missed.

So accuracy is a trap here. We chose the models with **Average Precision** instead. It checks if the real defaulters are at the top of the ranking, which is exactly what the call team needs.

Slide 5. Five models

We compared five models from the course on the 4,500 validation clients. The last column is the most concrete one: out of 450 calls, how many reach a future defaulter. Random calls would reach about **99**. Logistic Regression reaches 298, and the two best models, Random Forest and XGBoost, reach **310**.

These two are almost tied: 0.552 and 0.550 in Average Precision. When we ran the same code on two of our laptops, the XGBoost score changed a little. So the difference was only noise, and we kept the **Random Forest**, which gives the same result on every computer.

Slide 6. The final test

Then, and only then, we used the test set, one time. These are 4,500 clients the model had never seen. In its 450 highest scores, the Random Forest finds **317 real defaulters**. Random calls would find 99. So **7 calls out of 10** reach the right person, more than three times better than chance.

The ROC-AUC is 0.782 on the test set and 0.777 on validation. The results are very close, so the model does not overfit.

Lucas will now explain what this means in money.

Lucas slides 7 to 10, about 2 minutes 45

Slide 7. The cost of errors

The two errors do not cost the same. With our assumptions, missing a defaulter costs about **4,700 NT$**, that's New Taiwan dollars, and calling a good client costs **2,000**.

The curve shows the value created by the calls. With the model, it grows until about **650 calls**, so 450 calls is profitable. With random calls, the bank always loses money.

The costs are assumptions, so we tested **nine scenarios**. The model's list is profitable in all nine. In fact, a call only needs to save 2.3% of the bill to pay for itself, and we assumed 10%.

Slide 8. Recommendation

Our recommendation is a simple monthly rule. Call the **top 450**, with a reminder and an instalment plan. Send an automatic SMS to ranks 451 to 650, which is almost free. Do nothing for the others.

On the test month, this rule reaches 317 defaulters and brings about **1.8 million NT$**, around 51,000 euros, for 4,500 clients.

Slide 9. The tool (screenshots of our web page)

We also built a small web page that uses the model. The curve is the 4,500 clients, riskiest first, and the red part is the 450 the team calls.

Point at the left screenshot.

On the left, a client who was late in August and September. The client is ranked **827th**, so 377 places after the last call: we only send a reminder.

Point at the right screenshot.

On the right, a client who has been late for months: ranked **8th**, so the team calls this client this month.

Slide 10. Limits and next steps

To finish, four limits. First, the data is old and foreign: Taiwan, 2005. Another bank would retrain the model on its own clients and follow data-protection rules. Second, the model predicts who will default, not whether a call prevents it. The next step is an A/B test: call only half of the top 450 and compare the default rates. Third, the costs should be confirmed with the finance team. And finally, the model must be checked every month and retrained when it gets worse.

Thank you for your attention. We are happy to answer your questions.